

Thiết Kế Cơ Sở Dữ Liệu

Đề Tài: LivingDocs

Nhóm 1

2026

Mục lục

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Preface

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part I.

First half of the book

chapter 1.

Introduction

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1.1. Research question

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1.2. Answer to the question

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ullamco laboris nisi ut aliquip ex ea commodo consequat.² Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

1.3. Positioning in the literature

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1.4. A section

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1.4.1. A subsection

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1.4.2. Yet another subsection

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1.5. A final introduction section

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chapter 2.

Sections and content

part II.

Thiết kế & Cài đặt

chapter 3.

Thuật toán maping ER

3.1. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

3.1.1. Cú pháp Rút gọn (Logical Schema Syntax)

- `users` (user_id UUID PK, email VARCHAR(255), password_hash VARCHAR(255), first_name VARCHAR(100), last_name VARCHAR(100), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)[cite: 1]
- `workspaces` (workspace_id UUID PK, name VARCHAR(150), description TEXT, status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)[cite: 1]
- `roles` (role_id UUID PK, role_name VARCHAR(50), description TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)[cite: 1]
- `permissions` (permission_id UUID PK, permission_code VARCHAR(50), description TEXT, created_at TIMESTAMP)[cite: 1]
- `github_connections` (connection_id UUID PK, user_id UUID FK, github_user_id VARCHAR(50), github_username VARCHAR(100), access_token VARCHAR(500), created_at TIMESTAMP, updated_at TIMESTAMP)[cite: 1]

- `user_notification_preferences` (preference_id UUID PK, user_id UUID FK, channel VARCHAR(20))[cite: 1]
- `github_connection_scopes` (scope_id UUID PK, connection_id UUID FK, scope_value VARCHAR(50))[cite: 1]
- `workspace_members` (membership_id UUID PK, user_id UUID FK, workspace_id UUID FK, role_id UUID FK, joined_at TIMESTAMP, status VARCHAR(20))[cite: 1]
- `role_permissions` (role_permission_id UUID PK, role_id UUID FK, permission_id UUID FK, granted_at TIMESTAMP)[cite: 1]

3.1.2. Mã DDL SQL Hoàn Chỉnh (schema_member_1.sql)

```

1
2 CREATE TABLE users (
3     user_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
4     email            VARCHAR(255)  NOT NULL,
5     password_hash    VARCHAR(255)  NULL,
6     first_name       VARCHAR(100)   NULL,
7     last_name        VARCHAR(100)   NULL,
8     status           VARCHAR(20)    NOT NULL,
9     created_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
10    updated_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
11    deleted_at       TIMESTAMP      NULL,
12    CONSTRAINT pk_users PRIMARY KEY (user_id),
13    CONSTRAINT uq_users_email UNIQUE (email)
14 );
15
16
17 CREATE TABLE workspaces (
18     workspace_id     UUID          NOT NULL DEFAULT gen_random_uuid(),
19     name             VARCHAR(150)   NOT NULL,
20     description      TEXT           NULL,
21     status           VARCHAR(20)    NOT NULL,
22     created_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
23     updated_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
24     deleted_at       TIMESTAMP      NULL,
25     CONSTRAINT pk_workspaces PRIMARY KEY (workspace_id)
26 );
27
28
29 CREATE TABLE roles (
30     role_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
31     role_name        VARCHAR(50)   NOT NULL,
32     description      TEXT           NULL,

```

```

33     created_at      TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP ,
34     updated_at      TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP ,
35     CONSTRAINT pk_roles PRIMARY KEY (role_id)
36 );
37
38
39 CREATE TABLE permissions (
40     permission_id    UUID           NOT NULL DEFAULT gen_random_uuid(),
41     permission_code   VARCHAR(50)    NOT NULL ,
42     description       TEXT           NULL ,
43     created_at        TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP ,
44     CONSTRAINT pk_permissions PRIMARY KEY (permission_id),
45     CONSTRAINT uq_permissions_code UNIQUE (permission_code)
46 );
47
48
49 CREATE TABLE github_connections (
50     connection_id     UUID           NOT NULL DEFAULT gen_random_uuid(),
51     user_id            UUID           NOT NULL ,
52     github_user_id     VARCHAR(50)    NOT NULL ,
53     github_username     VARCHAR(100)   NOT NULL ,
54     access_token        VARCHAR(500)   NOT NULL ,
55     created_at          TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP ,
56     updated_at          TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP ,
57     CONSTRAINT pk_github_connections PRIMARY KEY (connection_id),
58     CONSTRAINT uq_github_connections_user UNIQUE (user_id),
59     CONSTRAINT fk_github_connections_user
60         FOREIGN KEY (user_id) REFERENCES users (user_id)
61         ON UPDATE CASCADE
62         ON DELETE CASCADE
63 );
64
65
66 CREATE TABLE user_notification_preferences (
67     preference_id      UUID           NOT NULL DEFAULT gen_random_uuid(),
68     user_id            UUID           NOT NULL ,
69     channel             VARCHAR(20)    NOT NULL ,
70     CONSTRAINT pk_user_notification_preferences PRIMARY KEY
71     (preference_id),
72     CONSTRAINT fk_user_notification_pref_user
73         FOREIGN KEY (user_id) REFERENCES users (user_id)
74         ON UPDATE CASCADE
75         ON DELETE CASCADE
76 );
77
78 CREATE TABLE github_connection_scopes (

```

```

79     scope_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
80     connection_id     UUID          NOT NULL,
81     scope_value        VARCHAR(50)   NOT NULL,
82     CONSTRAINT pk_github_connection_scopes PRIMARY KEY (scope_id),
83     CONSTRAINT fk_github_connection_scopes_conn
84         FOREIGN KEY (connection_id) REFERENCES github_connections
85         (connection_id)
86         ON UPDATE CASCADE
87         ON DELETE CASCADE
88 );
89 CREATE TABLE workspace_members (
90     membership_id     UUID          NOT NULL DEFAULT gen_random_uuid(),
91     user_id            UUID          NOT NULL,
92     workspace_id       UUID          NOT NULL,
93     role_id            UUID          NOT NULL,
94     joined_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
95     status              VARCHAR(20)  NOT NULL,
96     CONSTRAINT pk_workspace_members PRIMARY KEY (membership_id),
97
98     CONSTRAINT uq_workspace_members_user_workspace UNIQUE (user_id,
99     workspace_id),
100    CONSTRAINT fk_workspace_members_user
101        FOREIGN KEY (user_id) REFERENCES users (user_id)
102        ON UPDATE CASCADE
103        ON DELETE CASCADE,
104    CONSTRAINT fk_workspace_members_workspace
105        FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
106        ON UPDATE CASCADE
107        ON DELETE CASCADE,
108    CONSTRAINT fk_workspace_members_role
109        FOREIGN KEY (role_id) REFERENCES roles (role_id)
110        ON UPDATE CASCADE
111        ON DELETE RESTRICT
112 );
113
114 CREATE TABLE role_permissions (
115     role_permission_id UUID          NOT NULL DEFAULT gen_random_uuid(),
116     role_id             UUID          NOT NULL,
117     permission_id       UUID          NOT NULL,
118     granted_at           TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
119     CONSTRAINT pk_role_permissions PRIMARY KEY (role_permission_id),
120     CONSTRAINT uq_role_permissions_pair UNIQUE (role_id, permission_id),
121     CONSTRAINT fk_role_permissions_role
122         FOREIGN KEY (role_id) REFERENCES roles (role_id)
123         ON UPDATE CASCADE

```

```

124         ON DELETE CASCADE,
125     CONSTRAINT fk_role_permissions_permission
126         FOREIGN KEY (permission_id) REFERENCES permissions (permission_id)
127         ON UPDATE CASCADE
128         ON DELETE CASCADE
129 );
130
131
132 CREATE INDEX idx_github_connections_user_id ON github_connections(user_id);
133 CREATE INDEX idx_workspace_members_workspace_id ON
134     workspace_members(workspace_id);
135 CREATE INDEX idx_workspace_members_user_id ON workspace_members(user_id);
136 CREATE INDEX idx_role_permissions_role_id ON role_permissions(role_id);
137
138 CREATE OR REPLACE FUNCTION set_updated_at()
139 RETURNS TRIGGER AS $$
140 BEGIN
141     NEW.updated_at := CURRENT_TIMESTAMP;
142     RETURN NEW;
143 END;
144
145
146 CREATE TRIGGER trg_users_updated_at
147     BEFORE UPDATE ON users
148     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
149
150 CREATE TRIGGER trg_workspaces_updated_at
151     BEFORE UPDATE ON workspaces
152     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
153
154 CREATE TRIGGER trg_roles_updated_at
155     BEFORE UPDATE ON roles
156     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
157
158 CREATE TRIGGER trg_github_connections_updated_at
159     BEFORE UPDATE ON github_connections
160     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
161
162 GO

```


chapter 2.

Phân hệ Workspace & Extension (Thành viên 2)

2.1. Phân hệ Workspace & Extension

2.1.1. Cú pháp Rút gọn (Logical Schema Syntax)

- workspaces (workspace_id UUID PK, name VARCHAR(150), description TEXT, status VARCHAR(20), created_at DATETIMEOFFSET, updated_at DATETIMEOFFSET, deleted_at DATETIMEOFFSET)
- github_connections (connection_id UUID PK, workspace_id UUID FK, provider VARCHAR(50), account_name VARCHAR(100), access_token TEXT, sync_status VARCHAR(20), created_at DATETIME2, updated_at DATETIME2)
- github_connection_scopes (scope_id UUID PK, connection_id UUID FK, scope_value VARCHAR(100), created_at DATETIME2)
- repositories (repository_id UUID PK, workspace_id UUID FK, name VARCHAR(150), url VARCHAR(255), default_branch VARCHAR(50), created_at DATETIME2, updated_at DATETIME2)
- branches (branch_id UUID PK, repository_id UUID FK, name VARCHAR(100), is_protected BOOLEAN, created_at DATETIME2)

2.1.2. Mã DDL SQL Hoàn Chính (schema_member_workspace.sql)

```
1  -- 1. BANG WORKSPACES (Quan ly khong gian lam viec cua nguoi dung)
2  CREATE TABLE workspaces (
3      workspace_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
4      name VARCHAR(150) NOT NULL,
5      description TEXT NULL,
6      status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
7      created_at DATETIMEOFFSET DEFAULT SYSDATETIMEOFFSET(),
8      updated_at DATETIMEOFFSET DEFAULT SYSDATETIMEOFFSET(),
9      deleted_at DATETIMEOFFSET NULL,
10
11     -- Constraints
12     CONSTRAINT uk_workspace_id_name UNIQUE (workspace_id, name),
13     CONSTRAINT chk_workspace_status CHECK (status IN ('ACTIVE', 'ARCHIVED'))
14 );
15
16 -- 2. BANG GITHUB_CONNECTIONS (Ket noi tai khoan GitHub voi Workspace)
17 CREATE TABLE github_connections (
18     connection_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
19     workspace_id UNIQUEIDENTIFIER NOT NULL,
20     provider VARCHAR(50) NOT NULL DEFAULT 'GITHUB',
21     account_name VARCHAR(100) NOT NULL,
22     access_token TEXT NOT NULL,
23     sync_status VARCHAR(20) NOT NULL DEFAULT 'IDLE',
24     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
25     updated_at DATETIME2 NULL,
26
27     -- Constraints
28     CONSTRAINT uk_github_connection UNIQUE (connection_id, provider),
29     CONSTRAINT chk_sync_status CHECK (sync_status IN ('IDLE', 'SYNCING', 'SUCCESS', 'FAILED'))
30 );
31
32 -- 3. BANG GITHUB_CONNECTION_SCOPES (Pham vi quyen han cua ket noi GitHub)
33 CREATE TABLE github_connection_scopes (
34     scope_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
35     connection_id UNIQUEIDENTIFIER NOT NULL,
36     scope_value VARCHAR(100) NOT NULL,
37     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
38
39     CONSTRAINT uk_connection_scope UNIQUE (connection_id, scope_value)
40 );
41
42 -- 4. BANG REPOSITORIES (Kho luu tru ma nguon lien ket)
43 CREATE TABLE repositories (
```

```

44     repository_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
45     workspace_id UNIQUEIDENTIFIER NOT NULL,
46     name VARCHAR(150) NOT NULL,
47     url VARCHAR(255) NOT NULL,
48     default_branch VARCHAR(50) DEFAULT 'main',
49     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
50     updated_at DATETIME2 NULL,
51
52     CONSTRAINT uk_repository_name UNIQUE (repository_id, name)
53 );
54
55 -- 5. BANG BRANCHES (Cac nhanh lam viec trong Repository)
56 CREATE TABLE branches (
57     branch_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
58     repository_id UNIQUEIDENTIFIER NOT NULL,
59     name VARCHAR(100) NOT NULL,
60     is_protected BIT NOT NULL DEFAULT 0,
61     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
62
63     CONSTRAINT uk_branch_name UNIQUE (repository_id, name)
64 );
65 GO
66
67 -- =====
68 -- KHAI BAO CAC KHOA NGOAI THAM CHIEU (CROSS-DOMAIN & INTERNAL FKs)
69 -- =====
70 ALTER TABLE github_connections ADD CONSTRAINT
71     fk_github_connections_workspace
72     FOREIGN KEY (workspace_id) REFERENCES workspaces(workspace_id) ON
73     DELETE CASCADE;
74
75 ALTER TABLE github_connection_scopes ADD CONSTRAINT
76     fk_github_scopes_connection
77     FOREIGN KEY (connection_id) REFERENCES github_connections(connection_id)
78     ON DELETE CASCADE;
79
80 ALTER TABLE repositories ADD CONSTRAINT fk_repositories_workspace
81     FOREIGN KEY (workspace_id) REFERENCES workspaces(workspace_id) ON
82     DELETE CASCADE;
83
84 ALTER TABLE branches ADD CONSTRAINT fk_branches_repository
85     FOREIGN KEY (repository_id) REFERENCES repositories(repository_id) ON
86     DELETE CASCADE;
87
88 GO

```

2.2. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

2.2.1. Cú pháp Rút gọn (Logical Schema Syntax)

- reviews (review_id UUID PK, version_id UUID FK, user_id UUID FK, status VARCHAR(20), decision VARCHAR(20), summary_notes TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)
- review_comments (comment_id UUID PK, review_id UUID FK, user_id UUID FK, content TEXT, is_resolved BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP)
- approvals (approval_id UUID PK, version_id UUID FK, user_id UUID FK, decision VARCHAR(20), rejection_reason TEXT, approved_at TIMESTAMP)
- grounding_evidences (evidence_id UUID PK, version_id UUID FK, snippet_content TEXT, purpose VARCHAR(30), created_at TIMESTAMP)
- drift_alerts (alert_id UUID PK, evidence_id UUID FK, repository_id UUID FK, drift_type VARCHAR(20), status VARCHAR(20), detected_at TIMESTAMP)

2.2.2. Mã DDL SQL Hoàn Chỉnh (schema_member_5.sql)

```
1
2
3 CREATE TABLE reviews (
4     review_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
5     version_id UNIQUEIDENTIFIER NOT NULL,
6     user_id UNIQUEIDENTIFIER NOT NULL,
7     status VARCHAR(20) NOT NULL DEFAULT 'PENDING',
8     decision VARCHAR(20) NULL,
9     summary_notes NVARCHAR(MAX) NULL,
10    created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
11    updated_at DATETIME2 NULL,
12
13    CONSTRAINT chk_reviews_status CHECK (status IN ('PENDING', 'IN_PROGRESS', 'COMPLETED')),
14    CONSTRAINT chk_reviews_decision CHECK (decision IS NULL OR decision IN ('APPROVED', 'REJECTED', 'CHANGES_REQUESTED'))
15 );
16
17 CREATE TABLE review_comments (
18     comment_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
19     review_id UNIQUEIDENTIFIER NOT NULL,
20     user_id UNIQUEIDENTIFIER NOT NULL,
21     content NVARCHAR(MAX) NOT NULL,
```

```

22     is_resolved BIT NOT NULL DEFAULT 0,
23     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
24     updated_at DATETIME2 NULL,
25
26     CONSTRAINT fk_comments_review FOREIGN KEY (review_id)
27         REFERENCES reviews(review_id) ON DELETE CASCADE
28 );
29
30 CREATE TABLE approvals (
31     approval_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
32     version_id UNIQUEIDENTIFIER NOT NULL,
33     user_id UNIQUEIDENTIFIER NOT NULL,
34     decision VARCHAR(20) NOT NULL,
35     rejection_reason NVARCHAR(MAX) NULL,
36     approved_at DATETIME2 NOT NULL DEFAULT GETDATE(),
37
38     CONSTRAINT chk_approvals_decision CHECK (decision IN ('APPROVED', '
39     REJECTED'))
40 );
41
42 CREATE TABLE grounding_evidences (
43     evidence_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
44     version_id UNIQUEIDENTIFIER NOT NULL,
45     snippet_content NVARCHAR(MAX) NOT NULL,
46     purpose VARCHAR(30) NOT NULL DEFAULT 'HALLUCINATION_PREVENTION',
47     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
48
49     CONSTRAINT chk_evidences_purpose CHECK (purpose IN ('
50     HALLUCINATION_PREVENTION', 'CODE_REFERENCE', 'AST_VERIFICATION'))
51 );
52
53 CREATE TABLE drift_alerts (
54     alert_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
55     evidence_id UNIQUEIDENTIFIER NOT NULL,
56     repository_id UNIQUEIDENTIFIER NOT NULL,
57     drift_type VARCHAR(20) NOT NULL,
58     status VARCHAR(20) NOT NULL DEFAULT 'OPEN',
59     detected_at DATETIME2 NOT NULL DEFAULT GETDATE(),
60
61     CONSTRAINT fk_alerts_evidence FOREIGN KEY (evidence_id)
62         REFERENCES grounding_evidences(evidence_id) ON DELETE CASCADE,
63     CONSTRAINT chk_alerts_type CHECK (drift_type IN ('REFERENTIAL', '
64     SIGNATURE', 'SEMANTIC', 'OUTDATED')),
65     CONSTRAINT chk_alerts_status CHECK (status IN ('OPEN', 'RESOLVED', '
66     IGNORED'))
67 );

```

```

65
66
67 ALTER TABLE reviews ADD CONSTRAINT fk_reviews_version
68     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
69 ALTER TABLE reviews ADD CONSTRAINT fk_reviews_user
70     FOREIGN KEY (user_id) REFERENCES users(user_id);
71 ALTER TABLE review_comments ADD CONSTRAINT fk_comments_user
72     FOREIGN KEY (user_id) REFERENCES users(user_id);
73 ALTER TABLE approvals ADD CONSTRAINT fk_approvals_version
74     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
75 ALTER TABLE approvals ADD CONSTRAINT fk_approvals_user
76     FOREIGN KEY (user_id) REFERENCES users(user_id);
77 ALTER TABLE grounding_evidences ADD CONSTRAINT fk_evidences_version
78     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
79 ALTER TABLE drift_alerts ADD CONSTRAINT fk_alerts_repository
80     FOREIGN KEY (repository_id) REFERENCES repositories(repository_id);
81
82 GO

```

chapter 3.

Future research